

Title: Phase-type distributions for studying variability in resistive memories

Authors: C. Acal¹, J.E. Ruiz-Castro¹, A. M. Aguilera¹, F. Jiménez-Molinos², J.B. Roldán²

Address

¹Departamento de Estadística e Investigación Operativa and IEMath-GR. Universidad de Granada. Facultad de Ciencias. Campus de Fuentenueva s/n, 18071 GRANADA, Spain.

²Departamento de Electrónica y Tecnología de Computadores. Universidad de Granada. Facultad de Ciencias. Avd. Fuentenueva s/n, 18071 GRANADA, Spain.

Correspondence author e-mail: aaguiler@ugr.es

Abstract

A new statistical approach has been developed to analyze Resistive Random Access Memory (RRAM) variability. The stochastic nature of the physical processes behind the operation of resistive memories makes variability one of the key issues to solve from the industrial viewpoint of these new devices. The statistical features of variability have been usually studied making use of Weibull distribution. However, this probability distribution does not work correctly for some resistive memories, in particular for those based on the Ni/HfO₂/Si structure that has been employed in this work. A completely new approach based on phase-type modelling is proposed in this paper to characterize the randomness of resistive memories operation. An in-depth comparison with experimental results shows that the fitted phase-type distribution works better than the Weibull distribution and also helps to understand the physics of the resistive memories.

Key words

Resistive switching memory, RRAM, Conductive filaments, Reset process, Weibull distribution, Phase-type distributions, Statistical modelling, EM-algorithm

1. INTRODUCTION

In the context of applications for non-volatile memories, several emerging technologies are gaining momentum in the electronic industry. Among the new devices considered, both at industry and academia, RRAMs have shown an incomparable potential because they have good scalability, low power operation, fast speed and outstanding possibilities for fabrication in the current CMOS technology [1, 2, 3, 4, 5, 6].

The physics and internal properties of these devices have been studied by means of profound experimental studies [1, 7, 8, 9], and modelling and simulation studies [10, 11, 12, 13, 14, 15, 16]. A unified mathematical framework capable of describing and simulating the complex and interplaying electro-thermo-chemical processes that occur in this type of new emerging technologies in semiconductor device industry was recently developed [17]. Nevertheless, there are issues, such as variability, that have to be addressed prior to RRAM massive industrialization.

RRAMs operation is based on the stochastic nature of resistive switching (RS) processes that, in most cases, create (set process) and rupture (reset process) a conductive filament that changes drastically the device resistance [1, 5, 18]. There is a great need to analyse the statistics behind RRAM variability that is translated to different resistances, voltages and currents related to set and reset processes for each RS cycle (a cycle consist of a set process followed by a reset process) within a long series of cycles. Because of this, the choice of the right statistical model to describe the distribution of switching parameters (forming, set and reset voltages) is a critical requirement for RRAMs that ensures a robust design of the circuit and reliable data storage unit.